

Invoice Generator

Setup Guide & Blueprint

Complete invoice lifecycle automation with n8n, Google Sheets, Google Drive, and Gmail.

PHASE 1	PHASE 2	PHASE 3
CREATE • STORE • EMAIL	TRACK • REMIND • RECORD	PAY • CONFIRM • STOP

Purpose: Use this document as the setup guide, technical blueprint, and testing reference for the complete three-phase Invoice Generator.

Invoice Generator

A complete invoice lifecycle automation built with **n8n**, **Google**

Sheets, Google Drive, **and** Gmail.

The Invoice Generator automates an invoice from creation and delivery through due-date tracking, payment reminders, and payment confirmation.

■ What It Does

The agent is organized into three development phases:

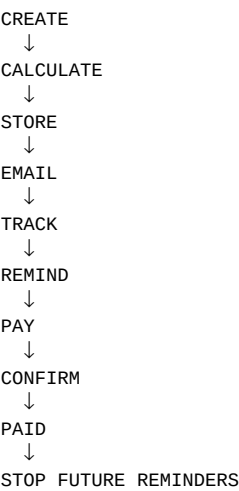
Phase Flow Purpose

Phase 1 Create → Store → Generate and deliver a Email professional invoice

Phase 2 Track → Remind → Monitor due dates and Record automate payment reminders

Phase 3 Pay → Confirm → Confirm payment and Stop automatically stop future reminders

Together, the phases create one complete invoice lifecycle:



■ Features

Phase 1 --- Create → Store → Email

- Receive invoice details through a webhook
- Validate invoice data
- Prepare and normalize invoice information
- Calculate:
 - Subtotal
 - Discount
 - Tax
 - Grand Total
 - Amount Due
- Generate a professional HTML invoice
- Convert the invoice into a PDF
- Store the PDF in Google Drive
- Record invoice information in Google Sheets
- Email the invoice to the customer

Phase 2 --- Track → Remind → Record

- Read invoice records from Google Sheets
- Check invoice payment status
- Calculate days until the due date
- Detect:
 - upcoming
 - due_soon
 - due_today
 - overdue
 - paid
- Automatically decide whether a reminder is required
- Send payment reminder emails
- Record the last reminder timestamp
- Maintain reminder count
- Skip reminders when no action is required

Phase 3 --- Pay → Confirm → Stop

- Receive payment confirmation through a webhook
- Validate payment confirmation data
- Find the invoice using its invoice number

- Mark the invoice as paid
- Set the outstanding amount to 0
- Return a payment confirmation response
- Reuse Phase 2's existing payment-status logic
- Automatically prevent future reminders for paid invoices

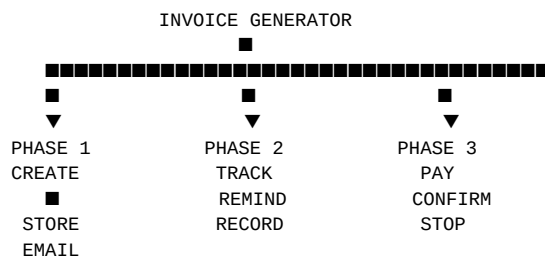
Important: Phase 3 is a payment-confirmation workflow. It does not

directly verify a transaction with a bank or payment gateway. A

payment provider integration can be added later.

■■ Architecture

The agent is split into three workflows so each responsibility remains clear and maintainable.



Phase 1

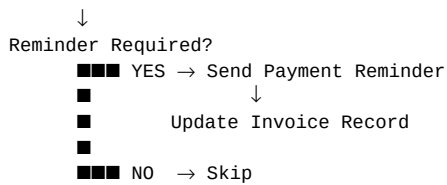
```

Invoice Request
↓
Validate Invoice Data
↓
Prepare Invoice Data
↓
Calculate Invoice
↓
Build Invoice Template
↓
Generate PDF
↓
Store Invoice
↓
Prepare Delivery Data
↓
Google Sheets Record
↓
Email Invoice
  
```

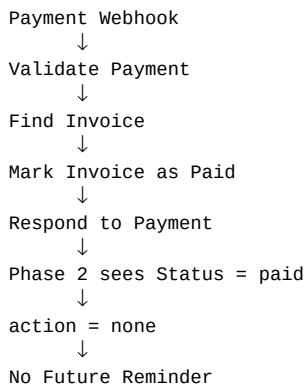
Phase 2

```

Schedule Trigger
↓
Get Invoice Records
↓
Check Invoice Status
  
```



Phase 3



■ Tech Stack

- **n8n** --- Workflow automation
- **Google Sheets** --- Invoice and reminder tracking
- **Google Drive** --- Invoice PDF storage
- **Gmail** --- Invoice delivery and payment reminders
- **PDF generation service** --- HTML invoice to PDF conversion
- **Postman** --- API/webhook testing

■ Google Sheets Structure

The invoice tracking sheet uses the following columns:

```

Invoice Number
Customer Name
Customer Email
Invoice Date
Due Date
Currency
Sub Total
Discount
Tax
Grand Total
amount Due
Status
Created At
Last Reminder
Reminder Count
  
```

Example

Invoice Number Customer Due Date Grand Total Amount Due Status

INV-2026-001 Chandu 2026-09-10 ■37,170 ■37,170 pending

After payment confirmation:

Invoice Number Amount Due Status

INV-2026-001 ■0 paid

■ Requirements

Before running the workflows, make sure you have:

- An n8n instance
 - Google account with access to:
 - Google Sheets
 - Google Drive
 - Gmail
 - A PDF conversion service/API configured for the PDF-generation step
 - Postman or another HTTP client for testing webhooks
 - The workflow JSON files from this repository
-

■ Credentials

The workflows require credentials for the services used by the nodes.

Configure the following in n8n:

Google Sheets

Used to:

- Create invoice records
- Read invoices
- Update payment/reminder information

Google Drive

Used to:

- Store generated invoice PDFs

Gmail

Used to:

- Send invoices
- Send payment reminders

PDF Service

Configure the API credentials required by the PDF generation node.

Never commit API keys, OAuth secrets, passwords, or private

credentials to this repository.

■ Installation

1. Install / open n8n

Use either:

- n8n Cloud
- A self-hosted n8n instance
- A local n8n installation

2. Import the workflows

Import the Phase 1, Phase 2, and Phase 3 workflow JSON files into n8n.

Recommended organization:

```
workflows/  
■■■ phase-1-invoice-generator.json  
■■■ phase-2-invoice-tracker.json  
■■■ phase-3-payment-confirmation.json
```

3. Configure credentials

Reconnect/configure the required Google Sheets, Google Drive, Gmail, and PDF-service credentials in n8n.

4. Configure the Google Sheet

Create the invoice tracking sheet using the column structure shown above.

5. Update workflow configuration

Check the following before activating the workflows:

- Google Sheet ID
 - Google Drive destination
 - Gmail sender account
 - PDF service credentials
 - Webhook paths
 - Email addresses
 - Time/date settings
-

■ Testing

Phase 1 --- Generate an Invoice

Send a POST request to the Phase 1 webhook.

Example:

```
{
  "business": {
    "name": "Shinka Solutions",
    "email": "billing@shinka.example",
    "phone": "+91 9876543210",
    "address": "Bengaluru, Karnataka, India",
    "taxId": "29ABCDE1234F1Z5"
  },
  "customer": {
    "name": "Chandu",
    "email": "chandouqiande@gmail.com",
    "phone": "+91 9123456789",
    "address": "Bengaluru, Karnataka, India"
  },
  "invoiceNumber": "INV-2026-001",
  "invoiceDate": "2026-08-26",
  "dueDate": "2026-09-10",
  "currency": "INR",
  "items": [
    {
      "description": "Website Development",
      "quantity": 1,
      "unitPrice": 25000
    },
    {
      "description": "Website Maintenance",
      "quantity": 2,
      "unitPrice": 5000
    }
  ],
  "discount": 10,
```



```
"tax": 18,  
"notes": "Thank you for your business."  
}
```

Expected calculation:

```
Subtotal      = ■35,000  
Discount      = ■3,500  
Taxable Amount = ■31,500  
Tax           = ■5,670  
Grand Total   = ■37,170  
Amount Due    = ■37,170
```

The workflow should then:

- Generate the invoice PDF
- Store it in Google Drive
- Record the invoice in Google Sheets
- Email the invoice to the customer

■ Phase 2 --- Test Reminders

The reminder workflow normally runs from its schedule trigger.

For testing, use an invoice with a due date close enough to trigger a reminder.

Example:

```
Due Date: 2026-09-01  
Current Date: 2026-08-30
```

Expected result:

```
invoiceState = due_soon  
action       = reminder
```

The workflow should:

```
Send Payment Reminder  
  ↓  
Update Last Reminder  
  ↓  
Increase Reminder Count
```

For an invoice that does not require a reminder:

```
action = none
```

the reminder branch should be skipped.

■ Phase 3 --- Confirm Payment

Send a POST request to the Phase 3 payment webhook.

Example:

```
{
  "invoiceNumber": "INV-2026-001",
  "paymentStatus": "paid"
}
```

Expected response:

```
{
  "success": true,
  "message": "Invoice marked as paid",
  "invoiceNumber": "INV-2026-001"
}
```

The Google Sheet should change from:

```
Status: pending
amount Due: ■37,170
```

to:

```
Status: paid
amount Due: ■0
```

■ Phase 2 + Phase 3 Integration Test

After marking an invoice as paid, run the Phase 2 tracking workflow again.

Because the invoice now has:

```
Status = paid
```

the tracking logic should produce:

```
invoiceState = paid
action       = none
```

The reminder condition should therefore evaluate to:

```
FALSE
```

Result:

No payment reminder is sent.

This verifies that payment confirmation successfully closes the reminder lifecycle.

■ Payment Validation

Phase 3 accepts only a confirmed paid status.

For example, this is valid:

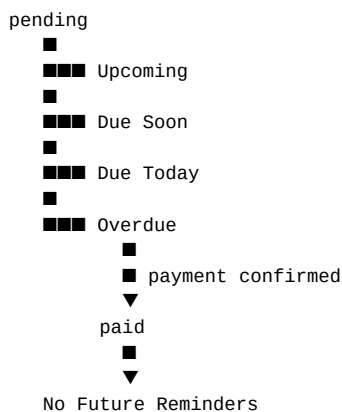
```
{
  "invoiceNumber": "INV-2026-001",
  "paymentStatus": "paid"
}
```

This should fail validation:

```
{
  "invoiceNumber": "INV-2026-001",
  "paymentStatus": "pending"
}
```

An invoice number is also required.

■ Invoice Status Lifecycle



■ Complete Example

A typical invoice moves through the system like this:

1. Customer invoice details received
- ↓
2. Invoice calculated
- ↓
3. PDF generated
- ↓
4. PDF stored in Google Drive
- ↓
5. Invoice recorded in Google Sheets
- ↓
6. Invoice emailed to customer
- ↓
7. Due date monitored
- ↓
8. Payment reminder sent when required

- ↓
9. Payment confirmed
- ↓
10. Invoice marked as paid
- ↓
11. Amount Due becomes 0
- ↓
12. Future reminders automatically skipped
-

■ Suggested Repository Structure

```
invoice-generator/  
■  
■■■ workflows/  
■ ■■■ phase-1-invoice-generator.json  
■ ■■■ phase-2-invoice-tracker.json  
■ ■■■ phase-3-payment-confirmation.json  
■  
■■■ docs/  
■ ■■■ setup.md  
■  
■■■ examples/  
■ ■■■ invoice-request.json  
■ ■■■ payment-confirmation.json  
■  
■■■ screenshots/  
■  
■■■ README.md  
■■■ LICENSE
```

Adjust the filenames to match the actual files in the repository.

■■ Important Notes

Payment verification

Phase 3 confirms a payment event supplied to the webhook. It does **not** independently verify that money was received.

For production use, connect the webhook to a trusted payment provider and verify its signed/event payload before marking an invoice as paid.

Webhook security

For production deployments, protect webhook endpoints using appropriate authentication/signature verification rather than exposing an unrestricted payment endpoint.

Google Sheets

The column names used by the workflows must match the configured sheet

fields.

Phase 2 scheduling

Phase 2 is designed around a scheduled workflow that periodically checks invoice records. A webhook response is therefore not part of the scheduled reminder execution itself.

■ Future Enhancements

The current version intentionally focuses on the core invoice lifecycle.

Possible future enhancements:

- Payment gateway integration
- Payment links
- Recurring invoices
- Customer management
- Revenue analytics
- Dashboard
- Multi-business support
- WhatsApp notifications
- Multi-currency improvements
- GST/e-invoicing integrations

These are intentionally outside the current three-phase implementation.

■ Version Summary

Phase 1

Create → Store → Email

Generate professional invoices, store them, record them, and deliver them automatically.

Phase 2

Track → Remind → Record

Monitor invoice due dates and automate payment reminders.

Phase 3

Pay → Confirm → Stop

Confirm payment, mark invoices as paid, clear the outstanding amount, and automatically stop future reminders.

■ One Complete Invoice Lifecycle

****Create → Calculate → Store → Email → Track → Remind → Pay → Confirm → Paid****

Built with **n8n + Google Sheets + Google Drive + Gmail**.

■ License

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For example:

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